FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE – GRADE 10 | SUB-STRAND 1.1: INTRODUCTION TO GENERAL SCIENCE	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS
This is your Final Explanation task. Over the past 8 lessons, you have been investigating one big question: How does a scientist know something is true when they cannot see it directly — and how is this power of General Science shaping life, careers, and the environment in Kenya? Now it is your turn to show everything you have learned by writing a thorough, evidence-based explanation. Read each section prompt carefully, study the exemplar response, then write your own answer in your own words. Your response should connect back to the maize farmer in Kisumu and the other evidence you gathered throughout the lesson sequence. Use scientific vocabulary you have learned, give specific Kenyan examples, and always support your claims with reasoning and evidence.

Section 1: The Problem of the Invisible — What Was the Farmer's Challenge?	
Describe the phenomenon that anchored this unit. What did the maize farmer in Kisumu observe, and why was her situation a scientific puzzle? In your answer, explain why 'seeing is believing' is NOT always a reliable rule in science, and identify at least TWO pieces of indirect evidence the farmer used to reach her conclusion.	The farmer noticed that maize plants along one edge of her shamba were wilting and turning yellow, even though she had watered and fertilised those plants exactly the same as the rest of the field. The puzzle was that she had never seen any insect on those plants — yet she concluded correctly that a soil pest was destroying the roots underground. This challenges the common idea that 'seeing is believing,' because many real causes in science are invisible to the naked eye. Scientists — and careful farmers — must rely on indirect evidence instead. The farmer used at least two pieces of indirect evidence: first, the pattern of the damage (only plants along one edge were affected, suggesting a localised underground cause rather than poor watering), and second, the timing (the wilting appeared at a season when certain soil pests are known to be active in the Kisumu region). By combining these clues — location, pattern, season, and the absence of any other visible cause — she built a logical case for a hidden cause she could not see directly. This is exactly what scientists do every day.

Section 2: What Is General Science and How Does It Help Us Know?	
Define General Science in your own words and explain its two main roles: (a) as a body of	General Science is both a collection of tested knowledge about the natural world AND a systematic way of investigating questions about that world. As a body of knowledge, it includes facts, concepts, and theories that scientists have built up over time — for example, knowledge about soil organisms, plant biology, and nutrient cycles. As a process, General Science gives us tools — like

knowledge and (b) as a process of investigation. Then explain the scientific practice of INFERENCE — what it is, how it is different from a random guess, and why it is one of the most powerful tools a scientist has. Use the farmer's story OR another Kenyan example to illustrate your explanation.	observing, measuring, experimenting, and inferring — that help us answer questions reliably even when we cannot see everything directly. One of the most powerful of these tools is inference. An inference is a logical conclusion drawn from evidence and prior knowledge — it is NOT a guess, because a guess is random. An inference is reasoned: you look at the clues you can observe, apply what you already know, and reach the most logical explanation. The maize farmer inferred a soil pest not because she guessed randomly, but because she knew that: (1) wilting along a single edge suggests a localised problem, (2) the fertiliser and water were equal across the field, and (3) soil pests like rootworms are common in her area during that season. Her inference was based on evidence plus knowledge — making it reliable, not lucky.

Section 3: Science in Action Across Kenya — Careers That Depend on Inference

General Science is not only for farmers. Describe at least THREE different Kenyan careers or fields where scientists and professionals must use indirect evidence and inference to reach conclusions. For each career, name the type of indirect evidence used and explain what conclusion it helps the professional reach. Show how all three examples connect back to the same core scientific skill the farmer used.	Across Kenya, many professionals use the same skill the Kisumu farmer used — reasoning from indirect evidence to an unseen truth. First, consider a clinical laboratory technologist at Kenyatta National Hospital in Nairobi. They cannot see the malaria parasite with the naked eye, but by examining a stained blood smear under a microscope, they detect the parasite's effects on red blood cells and infer that a patient has malaria. Second, consider an environmental scientist monitoring water quality near Lake Victoria. They cannot see dissolved nitrates or bacterial contamination directly, but by measuring chemical indicators in water samples, they infer dangerous levels of pollution and advise communities accordingly. Third, consider a doctor reading an X-ray in Kisumu County Hospital. They cannot see a fractured bone directly without surgery, but by studying the shadow and density patterns on the X-ray image, they infer the location and severity of the fracture. In every case, the professional observes something they CAN detect, applies scientific knowledge, and infers something they CANNOT see directly. This is the same logical process the maize farmer used — and it shows that inference is a universal scientific skill, not just an agricultural one.

Section 4: Science, the Environment, and Food Security in Kenya

Explain how the power of General Science — especially the ability to investigate invisible causes — is important for two of Kenya's most urgent challenges: (a) food security and (b) environmental protection. Give at least ONE specific example for each challenge. In your answer, explain what could go wrong if farmers, scientists, or policymakers did NOT use scientific reasoning and instead	Kenya's food security depends heavily on smallholder farmers growing crops like maize, beans, and sukuma wiki. Many of the threats to these crops — soil pests, fungal infections, nutrient deficiencies — are invisible to the naked eye. General Science gives farmers and agricultural scientists the tools to detect these hidden threats through soil testing, microscopy, and pattern analysis. For example, if every farmer in the Rift Valley waited to see a soil pest before acting, their entire crop could be destroyed by the time the damage became visible. Scientific inference allows them to act earlier, saving harvests and protecting food supplies. For environmental protection, consider air and water pollution around industrial areas near Mombasa or Nairobi. Many dangerous pollutants — heavy metals in rivers, carbon monoxide in air — are completely invisible and odourless. Without scientific instruments and inferential reasoning, communities would not know they were at risk until people became ill. Environmental scientists use chemical tests and data patterns to infer the presence and source of pollution long before it reaches crisis levels. In both cases, if people relied only on direct sight, they would miss the hidden causes of damage until it was too late. General Science gives Kenya the power to act on evidence, not just on what is visible.

relied only on what they could see with their eyes.	

Section 5: Your Conclusion — Answering the Driving Question	
Now answer the driving question directly and completely in your own words: How does a scientist know something is true when they cannot see it directly — and how is this power of General Science shaping life, careers, and the environment in Kenya? Your conclusion should: (1) summarise the key ideas from all four sections above, (2) explain what makes scientific inference RELIABLE rather than just a belief or an opinion, and (3) reflect on what YOU personally found most surprising or important about how science works.	A scientist knows something is true — even without seeing it directly — by gathering indirect evidence, applying tested knowledge, and using logical inference to reach the most well-supported conclusion. This is not guessing, and it is not mere belief. What makes scientific inference reliable is that it is based on observable evidence, connected to prior knowledge that has itself been tested, and open to revision if new evidence emerges. The maize farmer in Kisumu demonstrated this beautifully: she used pattern, location, timing, and the elimination of other causes to correctly identify an underground pest she never saw. Across Kenya, this same power shapes lives every day. Laboratory technologists diagnose malaria from blood cells; environmental scientists protect Lake Victoria from invisible pollutants; doctors read broken bones from X-ray shadows; agricultural researchers protect harvests from hidden soil threats. General Science matters because the most important things in nature — disease, pollution, soil health, cellular processes — are often invisible. A Kenya that trains its people to think scientifically, to reason from evidence, and to investigate hidden causes is a Kenya better equipped to protect its food security, its environment, and the health of its citizens. What I find most powerful about General Science is that it turns careful observation and logical thinking into a tool that is stronger than any single sense — it lets us see what our eyes alone never could.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Explanation of Indirect Evidence and Inference	Clearly and accurately defines inference, distinguishes it from guessing using specific reasoning, and applies it correctly to at least two detailed examples (including the Kisumu farmer). Demonstrates deep understanding of why indirect evidence is scientifically reliable.	Defines inference and distinguishes it from guessing. Applies the concept to the farmer's scenario and at least one other example. Explanation is mostly clear with minor gaps in reasoning.	Attempts to describe inference but confuses it with observation or guessing. Applies it to the farmer's scenario only, with limited reasoning. Explanation lacks precision or scientific vocabulary.
Use of Scientific Vocabulary and Concepts	Consistently and accurately uses scientific terms introduced in the unit (e.g., inference, indirect evidence, observation, hypothesis, evidence-based reasoning). Vocabulary is integrated naturally and enhances the explanation.	Uses most key scientific terms correctly. Occasional misuse or omission of vocabulary does not significantly weaken the explanation.	Uses few scientific terms, or uses them incorrectly. Explanation relies mostly on everyday language, reducing scientific precision.

Connection to Kenyan Contexts and Real-World Applications	Provides rich, specific Kenyan examples across at least three different careers or fields (e.g., medicine, environmental science, agriculture). Examples are accurate, detailed, and clearly connected to the core concept of inference from indirect evidence.	Provides Kenyan examples for at least two careers or fields. Examples are relevant and mostly accurate, though some connections to the core concept could be stronger.	Provides one Kenyan example or uses vague, generic examples not clearly tied to Kenya. Connection between the example and the concept of inference is unclear.
Response to the Driving Question	The final conclusion directly, completely, and confidently answers the driving question. It synthesises all four sections, explains what makes inference reliable (not just belief), and includes a thoughtful personal reflection grounded in the lesson content.	The final conclusion addresses the driving question and references ideas from most sections. Synthesis is present but may not fully integrate all parts. Personal reflection is included but may be brief or general.	The final conclusion partially addresses the driving question but misses key ideas or does not synthesise across sections. Personal reflection is absent or unconnected to the scientific content.
Clarity, Structure, and Evidence-Based Reasoning	Writing is clear, logically organised, and each claim is supported by specific evidence or reasoning. The explanation flows coherently from phenomenon to concept to application to conclusion. No unsupported assertions.	Writing is mostly clear and organised. Most claims are supported by evidence or reasoning, though one or two assertions may lack full support. The overall structure is easy to follow.	Writing is unclear in places or lacks logical organisation. Several claims are made without evidence or reasoning. The reader must infer connections the student has not made explicit.